

Aditya Malik

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TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, Go, TypeScript, R **AI/ML:** PyTorch, TensorFlow, NLP, Computer Vision, Scikit-learn, JAX, Model Evaluation **Infrastructure:** Docker, Linux, AWS (Lambda, S3, CDK), GCP, CI/CD, Git **Backend & Data:** FastAPI, Flask, PostgreSQL, MongoDB, Redis, Celery/RabbitMQ, WebSockets **Research:** Federated Learning, Differential Privacy, Edge AI, Conversational AI, 3D CNNs

EXPERIENCE

AI Research Intern - Alpha AI | Remote | Jun 2021 – Jul 2021

- Developed an end-to-end NLP pipeline for text summarization and neural voice cloning, training sequence-to-sequence and vocoder models with PyTorch - directly applicable to conversational AI model development.
- Built and evaluated ML models for autonomous driving in the CARLA simulator, applying computer vision techniques for real-time perception and decision-making.
- Delivered research outputs spanning NLP, computer vision, and reinforcement learning - demonstrating breadth across the ML research-to-implementation spectrum.

Software Developer - 3 Pillar Global | Remote | Dec 2021 – Jul 2022

- Engineered high-volume REST APIs (.NET, C#) serving 1M+ students across 1,000+ institutions - designed for reliability, observability, and performance at scale.
- Built automated unit test suites increasing coverage by 25%, reducing P1 production incidents by 30% through systematic monitoring and regression prevention.
- Drove CI/CD pipeline improvements that accelerated deployment velocity for a multi-tenant SaaS education platform.

Full Stack Developer - KCC Buildcon Pvt. Ltd. | Gurugram, India | Oct 2022 – Apr 2023

- Architected and delivered a corporate web platform for government clients (NHAI), integrating front-end and backend systems with PostgreSQL.
- Shipped 10+ features across the full stack, accelerating release cycles by 20% through iterative delivery and stakeholder collaboration.

PROJECTS

Edge-Centric Federated Learning for Autonomous Vehicles (MS Thesis)

- Designed a three-tier federated learning architecture deploying T-GCN models on Raspberry Pi (CAV), Jetson Xavier (RSU), and RTX A6000 (Coordinator) - a research-to-deployment pipeline bridging ML experimentation and real hardware.
- Achieved 50% lower inference latency and 6x bandwidth savings vs. cloud baselines while capping accuracy degradation under 2% with differential privacy - demonstrating rigorous model evaluation and production-readiness metrics.
- Integrated 5G NR-V2X communication patterns and smart-city data flows, outperforming centralized cloud benchmarks by 18%.

Arachnin - Production-Grade Web Retrieval Service for RAG

- Built a domain-crawling service that recursively scrapes websites, extracts clean article content, and produces structured LLM-ready output for downstream RAG pipelines - directly supporting conversational AI use cases.
- Architected an async distributed job processing system with FastAPI, Celery + RabbitMQ, Redis caching, and PostgreSQL - designed for horizontal scaling and production observability.
- Containerised the full stack with Docker Compose, implementing health checks and structured logging across all services.

Human Activity Recognition Using Deep Learning

- Designed and trained a 3D CNN architecture for video-based activity classification, implementing a full data processing pipeline handling the Kinetics-400 dataset under memory constraints.
- Evaluated pretrained backbone transfer learning against training from scratch, achieving ~80–90% classification accuracy with systematic ablation studies - demonstrating rigorous model evaluation methodology.
- Built reproducible training pipelines with configurable hyperparameters, checkpointing, and metric logging.

Aether Protocol - Decentralised P2P Content Delivery Network

- Designed a P2P CDN with a concurrent Go daemon handling cryptographic verification and network routing, paired with a React/Wails desktop client.
- Implemented Kademlia DHT for peer discovery and GossipSub for low-latency swarm distribution using libp2p.
- Built a content-addressed storage system with Ed25519 signatures, AES-GCM encryption, and a Web-of-Trust reputation model.

EDUCATION

MS Data Science - Deakin University, Melbourne | Jul 2025 | Distinction **B.Tech CSE (AI & ML Specialization)** - UPES, Dehradun, India | Jun 2022 | GPA: 7.81/10